

Adding modal DOFs to a fixed-budget XPBD host?

a diagnostic + operating guide — not a solver ranking, not a governor method

1 Can you change the contact row?

Make its modal weight stiffness-aware:

implicit $(M+hD+h^2K)^{-1}$, not $1/M$.

within-host control: 8/24→0/24 overdraws, no host swap

2 Stuck with the XPBD shared row?

Audit it: prefer iterations over substeps;
band-limit the basis.

32×1 holds ($R=0.30$); 4×8 overdraws (+481 J);
band-limiting still leaves 6/8 cells over

3 Need a hard energy guarantee?

The cumulative storage bound contains
modal energy — last resort.

holds in 90/90 cells, but still leaves up to 17.8 mm
penetration + a worse trajectory. Containment, not a fix.

The governor is a safety net, not the product.